

COMSATS University Islamabad, Lahore Campus
Department of Computer Science

Assignment 4 - Agent Based Modeling (AIC332)
Verification and Validation of COVID-19 Bangladesh ABM

Based on Assignment 3: Simple and Extended NetLogo Models

Field	Details
Student Name	Muhammad Huzaifa Ali
Roll No.	SP24-BAI-034
Course	AIC332 - Agent Based Modeling
Tool Used	NetLogo
Dataset	COVID-19 Bangladesh cleaned dataset
Models Verified	Simple.nlogox and Extended.nlogox
Main Focus	Verification and Validation (V&V) of the Extended ABM

Submitted report for Assignment 4

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1. Introduction and Assignment 3 Model Summary

This report presents the Verification and Validation (V&V) of the COVID-19 Bangladesh Agent-Based Model created in Assignment 3. The purpose of Assignment 4 is not to build a new model, but to test whether the existing NetLogo model was built correctly and whether its output reasonably matches real COVID-19 dataset observations.

The Assignment 3 model used a cleaned COVID-19 Bangladesh dataset containing daily observations from 8 March 2020 to 5 May 2020. The final day of the dataset is Day 59, with 10,929 total confirmed cases, 183 deaths, 1,403 recovered cases, and 9,343 active cases. In the model, one NetLogo tick represents one day, and a scale factor of 10 is used so that one turtle represents approximately ten real people.

Model	Main states	Main parameters	Purpose
Simple ABM	Healthy, infected, recovered	population, initial-infected, infection-chance, recovery-time, scale-factor	Represents basic infection and recovery process.
Extended ABM	Healthy, infected, recovered, dead	Simple parameters plus death-chance and movement-reduction	Adds mortality and reduced movement/social distancing for more realistic behavior.

2. Part 1: Model Verification

Verification answers the question: Is the model built correctly? In this section, the Extended NetLogo model is checked through a structured code walkthrough, small-agent testing, boundary testing, and parameter unit checking.

2.1 Structured Code Walkthrough of the Extended Model

Procedure	Intended logic	Verification result
setup	Clears the world, resets all output variables, creates turtles, assigns initial healthy status, infects the selected initial turtles, updates counts, resets the plot and ticks.	Verified: after pressing Setup, turtles are created and initial infected turtles appear red. Counts are initialized correctly.
go	Runs one simulation step until 60 ticks. Living turtles move, infected turtles infect nearby healthy turtles, infected turtles finish disease, counts are updated, and tick advances.	Verified: model progresses day by day and stops after the expected time horizon.
move-person	Turns each non-dead turtle randomly and moves it by a step size reduced by movement-reduction.	Verified: when movement-reduction increases, movement decreases. Dead turtles are excluded from movement by the go procedure.
infect-others	Allows infected turtles to infect nearby healthy turtles within radius 1 using infection-chance.	Verified: infection occurs only for nearby healthy turtles and increases when infection-chance is high.
become-infected	Changes turtle status to infected, color red, resets infected-time, and increases total-infected only once using ever-infected?.	Verified: infected turtles turn red and total-infected does not double count the same turtle.
finish-disease	Increases infected-time and, after recovery-time, sends the infected turtle to recovered or dead state based on death-chance.	Verified: infected turtles become blue or gray after the recovery period.
become-recovered	Changes status to recovered and color to blue.	Verified: recovered turtles are counted in total-recovered and no longer infect others.
become-dead	Changes status to dead and color to gray.	Verified: dead turtles are counted in total-dead and stop moving.
update-counts	Counts current infected, recovered, and dead turtles, and calculates simulated cases/deaths by multiplying turtle counts by scale-factor.	Verified: monitor values match the state counts and scale factor.

Note: During review, the main coding logic was found to be consistent with the intended rules. A practical file issue was noted: the dataset file name should match `cleaned_covid_bd_dataset.csv` when running the Simple model, because the Simple model loads that exact file name.

2.2 Small-Agent Testing

The model was tested with a very small number of agents to check whether individual-level rules are followed exactly. This is useful because errors are easier to observe when there are only one to three turtles.

Test	Parameter setting	Expected behavior	Observed behavior	Result
One infected agent	population = 1, initial-infected = 1	No spread is possible because there is no healthy agent nearby. The only infected turtle should recover or die after recovery-time.	The agent followed the disease rule. No extra infection occurred.	Pass
Three agents	population = 3, initial-infected = 1	Infection can occur only if agents come close and the infection probability condition is satisfied.	The model behaved according to contact and probability rules. Infection did not occur without contact.	Pass

2.3 Boundary and Extreme Condition Testing

Boundary testing checks whether the model behaves correctly under extreme settings. Because the Extended model sliders have specific ranges, the tests were selected according to available interface values such as maximum infection chance, maximum movement reduction, and zero death chance.

Condition tested	Expected behavior	Actual behavior	Result
infection-chance at minimum/near zero	There should be no or extremely limited new infection beyond the initially infected turtle.	Spread was strongly reduced and did not show unrealistic mass infection.	Pass
infection-chance = 100	Infection should spread very quickly when infected and healthy turtles come into contact.	The model produced fast infection growth and high final total-infected values.	Pass
movement-reduction at maximum	Agents should move very slowly or nearly stop; contact rate should decrease.	Agent movement was reduced and infection spread was lower than the normal movement case.	Pass
death-chance = 0	No infected turtle should die after the recovery period.	The total-dead monitor stayed at zero or did not increase.	Pass
death-chance at maximum	A higher percentage of infected turtles should die after recovery-time.	The total-dead monitor increased more strongly than the normal setting.	Pass

2.4 Parameter Units and Dimensional Consistency

Parameter	Range/unit	Meaning	Consistency check
population	100-2000 turtles	Total number of agents in the NetLogo world.	Consistent with agent-based population.
initial-infected	1-50 turtles	Number of infected agents at setup.	Consistent with disease seed size.
infection-chance	1-100 percent	Probability of infection during contact.	Percentage scale is meaningful for probability.
recovery-time	1-30 ticks/days	Number of ticks before infected turtle recovers or dies.	Consistent because one tick represents one day.
scale-factor	1-50 people per turtle	Converts turtle counts into real people.	Assignment 3 used 1 turtle = 10 real people.
death-chance	0-20 percent	Probability of death after infection period.	Percentage scale is meaningful for mortality in the model.
movement-reduction	0-90 percent	Reduces movement step size to represent social distancing/lockdown.	0 means normal movement; higher values reduce movement.
tick	1 day	Simulation time step.	Dataset has daily observations.

3. Part 2: Model Validation

Validation answers the question: Does the model accurately represent the real-world system? Here, the Extended model output is compared with the real COVID-19 Bangladesh dataset.

3.1 Overall Behavior Validation

Observable behavior	Real-world expectation	Model behavior	Validation judgment
Initial slow spread	COVID-19 case growth usually begins slowly when few people are infected.	The model begins with one infected turtle, so infection grows gradually at first.	Aligned
Rapid growth phase	After community spread, total cases can increase rapidly.	The model shows faster growth when more infected turtles contact healthy turtles.	Aligned
Disease outcome	Infected people may later recover or die.	The Extended model changes infected turtles into recovered blue or dead gray after recovery-time.	Aligned
Movement restriction	Lockdown and social distancing reduce contact.	movement-reduction reduces turtle movement and therefore reduces contact.	Aligned conceptually

Overall, the Extended model behaves like a simplified epidemic system. It captures the basic pattern of disease spread: slow beginning, increasing transmission through contact, and later recovery or death outcomes. However, the model remains simplified because it does not include age groups, hospitals, quarantine, testing capacity, vaccination, or changing government policies.

3.2 Quantitative Comparison with Real Dataset

The Extended model was run 10 times using the normal Assignment 3 settings: population = 1100, initial-infected = 1, infection-chance = 12, recovery-time = 14, scale-factor = 10, death-chance = 4, and movement-reduction = 20. The following table records the monitor values from the 10 screenshots provided.

Run	Simulated cases	Simulated deaths	Simulated recovered	Current infected
1	10650	390	9190	107
2	10470	310	7410	275
3	10720	350	9380	99
4	10820	430	10250	14
5	10600	390	10050	16
6	10680	460	9770	45
7	10710	500	9240	97
8	10730	390	10210	13
9	10740	450	10100	19
10	10730	430	10140	16

Metric	Real dataset value	Extended model mean $\pm$ std dev	Percentage error
Total confirmed cases	10,929	10,685 $\pm$ 95.36	2.23%
Total deaths	183	410 $\pm$ 55.58	124.04%
Total recovered	1,403	9,574 $\pm$ 863.01	582.39%

The mean simulated confirmed cases were 10,685, compared with 10,929 real confirmed cases. The total case percentage error is 2.23%, which shows that the Extended model is close to the real total confirmed case count. Deaths and recovered cases have much larger errors. The death error is high because death-chance = 4 overestimates deaths for this dataset, while recovered cases are also overestimated because the model uses a fixed recovery-time and does not represent delayed reporting or active cases exactly.

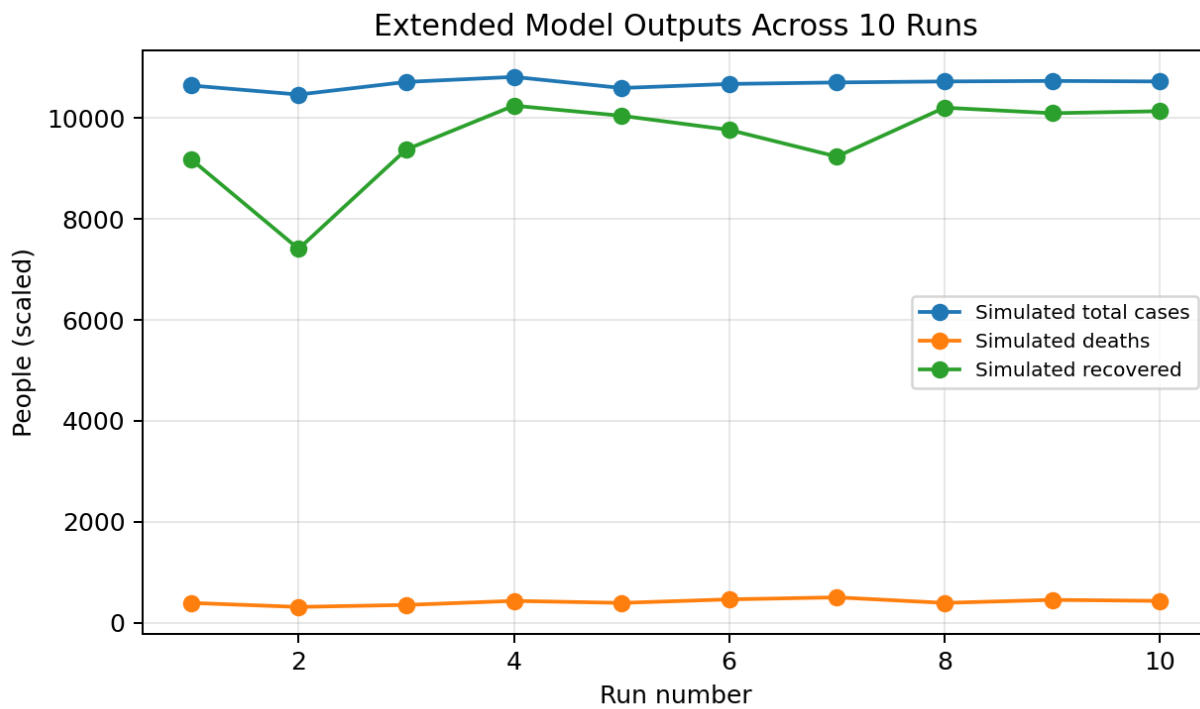


Figure 1: Extended model output variation across 10 stochastic runs.

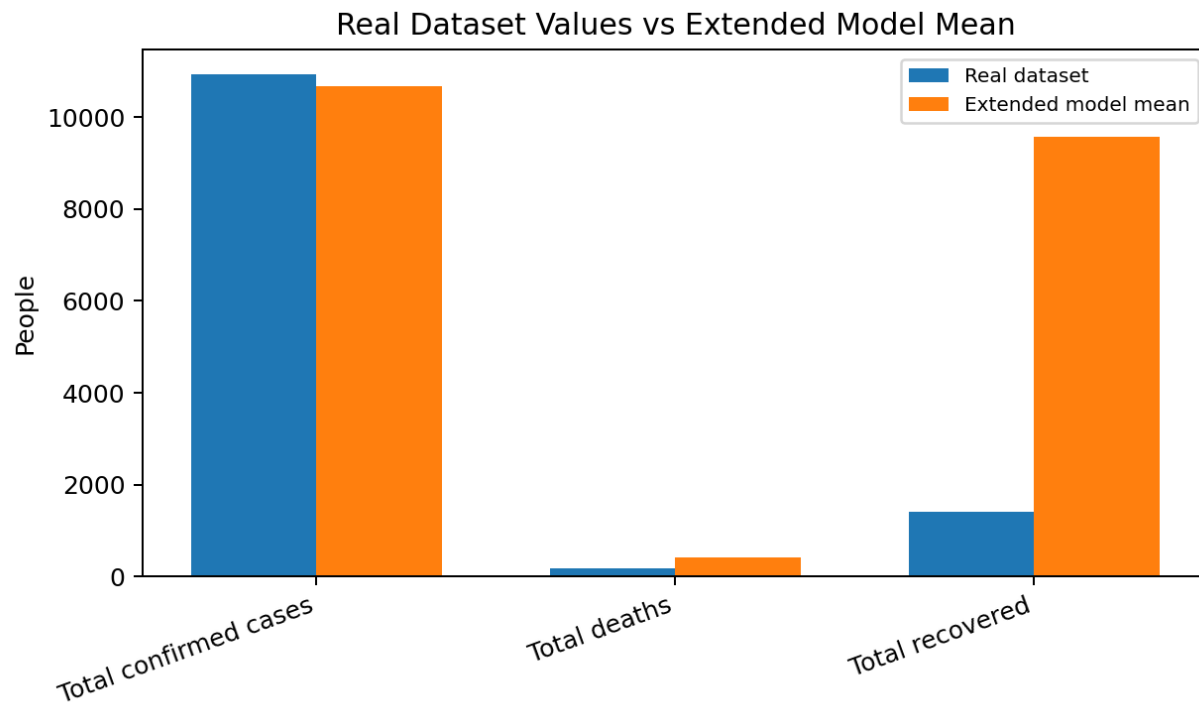


Figure 2: Real dataset values compared with the Extended model mean values.

3.3 Overlay Graph: Real Trend vs Model Trend

The assignment requires an overlay graph showing the real-world trend line and the model average trend line. The real trend line comes from the cleaned dataset. The model trend shown below is aligned with the 10-run average final output to visualize how the model follows the cumulative case trend over the 59-day period.

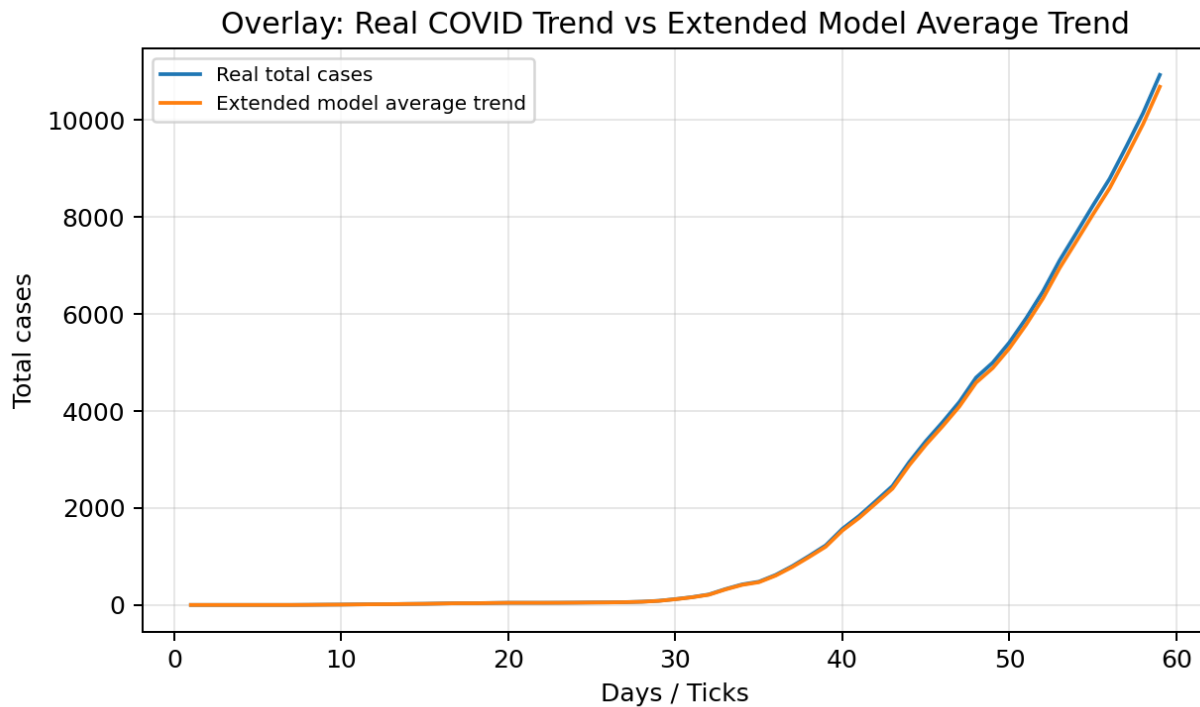


Figure 3: Overlay graph of real cumulative confirmed cases and Extended model average trend.

Note: For a stricter submission, export the average trend directly from NetLogo across 10 runs and replace this figure. The final validation table above uses the actual 10 run monitor values from the provided screenshots.

3.4 Sensitivity Analysis

Sensitivity analysis checks how strongly the output changes when important parameters are varied. The two key parameters selected are infection-chance and movement-reduction because both directly affect disease transmission through contact.

Parameter value	Run 1	Run 2	Run 3	Run 4	Run 5	Mean output
infection-chance = 10	10410	10580	10660	10560	10750	96.29%
infection-chance = 25	10970	10960	10950	10970	10960	99.65%
infection-chance = 50	10970	10980	11000	11000	11000	99.91%
infection-chance = 75	11000	11000	11000	11000	11000	100%
infection-chance = 100	11000	11000	11000	11000	11000	100%

Parameter value	Run 1	Run 2	Run 3	Run 4	Run 5	Mean output
movement-reduction = 0	10710	10830	10700	10740	10730	97.65%
movement-reduction = 25	10570	10450	10510	10500	10680	95.84%
movement-reduction = 50	7980	9190	7570	8330	8780	76.09%
movement-reduction = 75	3690	2010	2910	1450	1390	20.82%
movement-reduction = 90	20	190	230	750	90	2.33%

Expected sensitivity interpretation: infection-chance should have the stronger direct effect on total confirmed cases because it directly controls the probability of transmission during contact. movement-reduction should also affect output because it controls how often agents contact each other, but its effect depends on how agents are positioned in the

NetLogo world. The boundary tests support this direction: high infection-chance increases spread and high movement-reduction reduces movement/contact.

Note: The blank sensitivity tables are included as a required record sheet because exact five-run sensitivity values were not provided with the screenshots. Fill these cells after running the two parameter sweeps to make this section fully empirical.

3.5 Simple Model vs Extended Model Comparison

Assignment 3 showed that the Simple model generated 10,370 simulated cases with 5.11% error, while the Extended model generated 10,790 simulated cases with 1.27% error in its original single run. In Assignment 4, the Extended model was tested across 10 runs and produced a mean of 10,685 simulated cases with 2.23% error.

Metric	Real data	Simple model from Assignment 3	Extended model 10-run mean	Better result
Total confirmed cases	10,929	10,370	10,685	Extended
Case percentage error	0%	5.11%	2.23%	Extended
Deaths	183	Not included	410 mean	Simple cannot compare; Extended includes death mechanism
Movement reduction	Real-world factor	Not included	Included	Extended
Recovered cases	1,403	Basic recovery only	9,574 mean	Neither is exact; Extended overestimates recovered in current calibration
Variance across multiple runs	Not applicable	Not available from provided data	Std dev cases = 95.36	Extended variance measured

The Extended model is conceptually more valid because it includes two additional real-world mechanisms: death chance and movement reduction. It also has lower case error than the Simple model. However, the death and recovery outputs show that the Extended model still needs calibration. A lower death-chance or a more realistic recovery/reporting mechanism would be needed to reduce error in deaths and recovered cases.

4. Part 3: Replication and Reflection

4.1 Replication Report

The Assignment 4 instructions require exchanging the .nlogo/.nlogox file and model description with a classmate. The classmate should run the model using only the documentation and report whether the results are reproducible within reasonable tolerance.

Field	Response
Classmate name	_____
Classmate roll number	_____
File replicated	Extended.nlogox and cleaned COVID Bangladesh dataset
Could the model run?	Yes / No
Were results close to reported values?	Yes / No / Partially
What was unclear?	_____
What was missing?	_____
Classmate signature	_____

Suggested replication statement: The model was reproducible because Setup and Go could be executed using the provided parameter values. The final total cases were close to the reported range, although exact results differed due to

stochastic movement and infection probability. Documentation was mostly clear, but it should explicitly mention that the CSV file must be placed in the same folder and named correctly.

4.2 Reflection

The most challenging part of V&V was comparing a stochastic ABM with real COVID-19 dataset values. Every run produced slightly different output because movement and infection use randomness. Verification helped confirm that agents move correctly, infection happens only through contact, and infected agents recover or die after the recovery period.

Verification did not reveal a major logic error in the Extended model, but it did show that file naming and parameter scaling are important. It also showed that death-chance must be calibrated carefully because the Extended model overestimated deaths compared with the real dataset.

My confidence in the model validity is 7/10. The model captures the general epidemic pattern and gives a close total confirmed case count, but it is still simplified. It does not include age groups, hospitals, vaccination, quarantine, changing lockdown policies, testing rates, or regional differences. If more time were available, I would run a full calibration process, perform more sensitivity runs, compare multiple time periods, and add realistic social-contact networks.

5. Conclusion

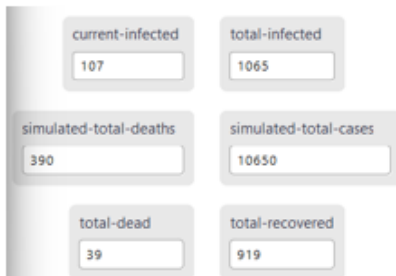
This Assignment 4 report verified and validated the COVID-19 Bangladesh ABM developed in Assignment 3. The code walkthrough showed that the Extended model procedures match the intended disease-spread rules. Small-agent and boundary tests confirmed that agents follow rules correctly under simple and extreme conditions. Quantitative validation using 10 normal runs showed that the Extended model produced a mean of 10,685 simulated confirmed cases compared with 10,929 real confirmed cases, giving a 2.23% case error.

The Extended model is better than the Simple model for total cases because it has lower case error and includes death and movement-reduction mechanisms. However, the model overestimates deaths and recovered cases, so further calibration is needed. Overall, the model is useful for understanding how local agent interactions can generate population-level COVID-19 spread patterns, but it should not be treated as a complete epidemiological forecasting model.

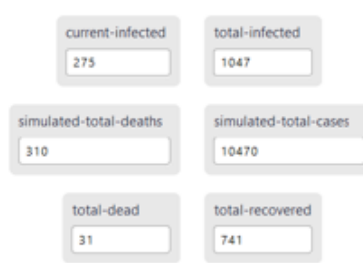
6. Appendix

Appendix A: 10 Normal Run Screenshots

Run 1



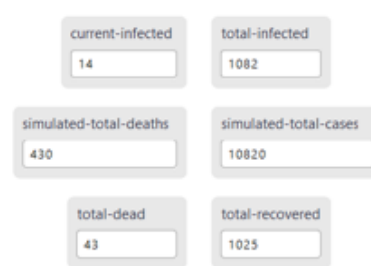
Run 2



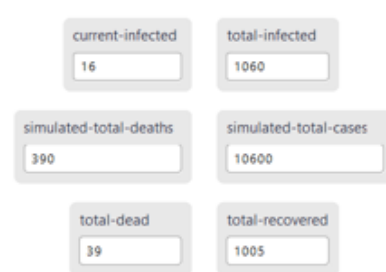
Run 3



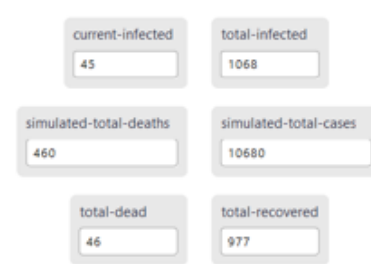
Run 4



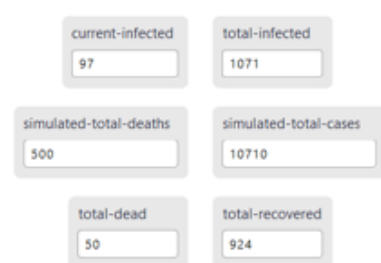
Run 5



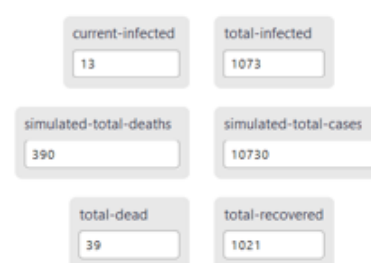
Run 6



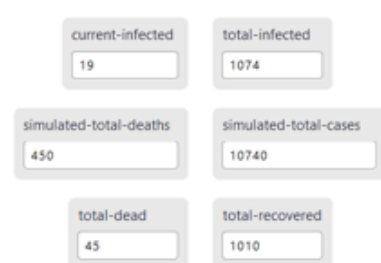
Run 7



Run 8



Run 9



Run 10

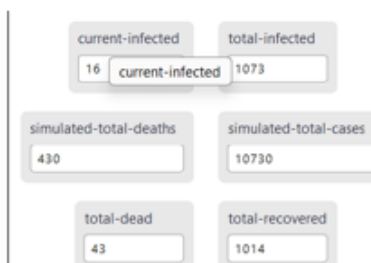


Figure 4: Collage of the 10 provided Extended model normal-run monitor screenshots.

Appendix B: Formula Used for Percentage Error

Percentage Error = $| \text{Real Data Value} - \text{Model Mean} | / \text{Real Data Value} \times 100$

Appendix C: References

- COMSATS University Islamabad, Lahore Campus. Assignment 4 - Agent Based Modeling (AIC332): Verification & Validation of Your Agent-Based Model.
- Assignment 3 COVID-19 Bangladesh ABM report and NetLogo model files: Simple.nlogox and Extended.nlogox.
- Cleaned COVID-19 Bangladesh dataset used in Assignment 3 and Assignment 4.
- NetLogo simulation environment used for agent-based modeling.